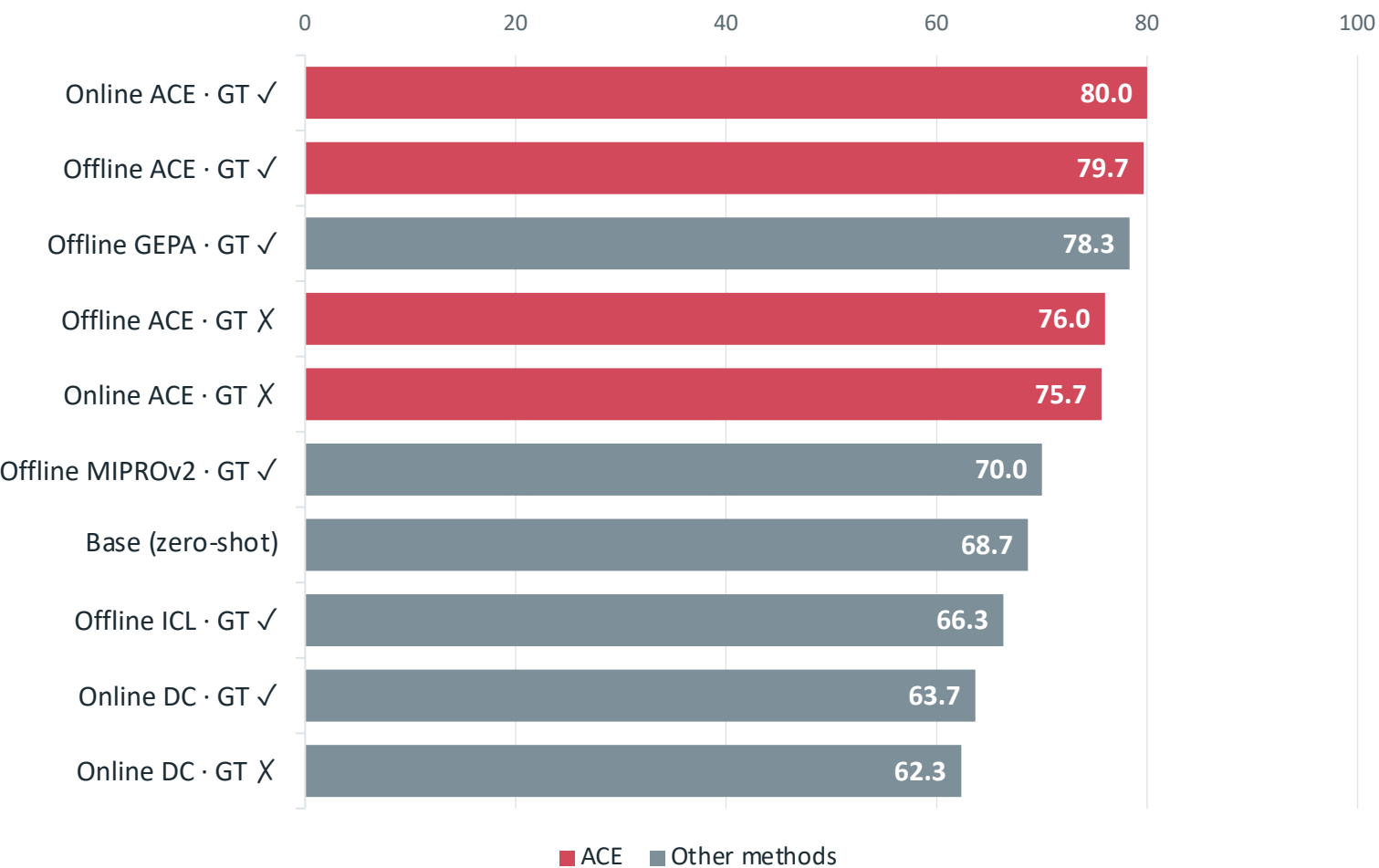


Dreaddit: ACE and GEPA beat the base model by 7–11 points

Qwen3-8B (Q4_K_M) · 100 test posts · mean over seeds 42, 43, 44 · binary stress accuracy (%)



+11.3 pp

Online ACE, GT ✓
80.0 vs. Base 68.7

+9.7 pp

Offline GEPA, GT ✓
78.3, the best prompt optimizer

−6.3 pp

Online DC, GT ✗
62.3, the lowest condition

pp = percentage points against the matched Base mean. Internal working results from the v5 server panel; not manuscript evidence.

Dreaddit: full panel and what to keep in mind

Mean $\pm$ sample SD over three seeds; Δ Acc against the matched Base mean

Condition	GT	Acc %	Macro-F1 %	Conformability %	Δ Acc
Base (zero-shot)	—	68.67 $\pm$ 2.31	67.55 $\pm$ 1.60	91.33 $\pm$ 2.08	0.00
Offline ICL	✓	66.33 $\pm$ 0.58	63.91 $\pm$ 0.77	83.33 $\pm$ 2.52	−2.33
Offline MIPROv2	✓	70.00 $\pm$ 4.58	68.63 $\pm$ 5.44	97.67 $\pm$ 1.53	+1.33
Offline GEPA	✓	78.33 $\pm$ 2.08	78.33 $\pm$ 2.08	92.33 $\pm$ 2.08	+9.67
Offline ACE	✓	79.67 $\pm$ 2.89	78.90 $\pm$ 3.28	76.00 $\pm$ 19.29	+11.00
Offline ACE	X	76.00 $\pm$ 6.24	75.25 $\pm$ 7.11	93.67 $\pm$ 4.73	+7.33
Online DC (CU)	✓	63.67 $\pm$ 0.58	60.49 $\pm$ 0.77	94.67 $\pm$ 3.21	−5.00
Online DC (CU)	X	62.33 $\pm$ 1.53	58.83 $\pm$ 2.32	91.33 $\pm$ 3.51	−6.33
Online ACE	✓	80.00 $\pm$ 2.65	79.92 $\pm$ 2.74	89.00 $\pm$ 1.00	+11.33
Online ACE	X	75.67 $\pm$ 5.51	75.06 $\pm$ 5.98	90.67 $\pm$ 7.57	+7.00

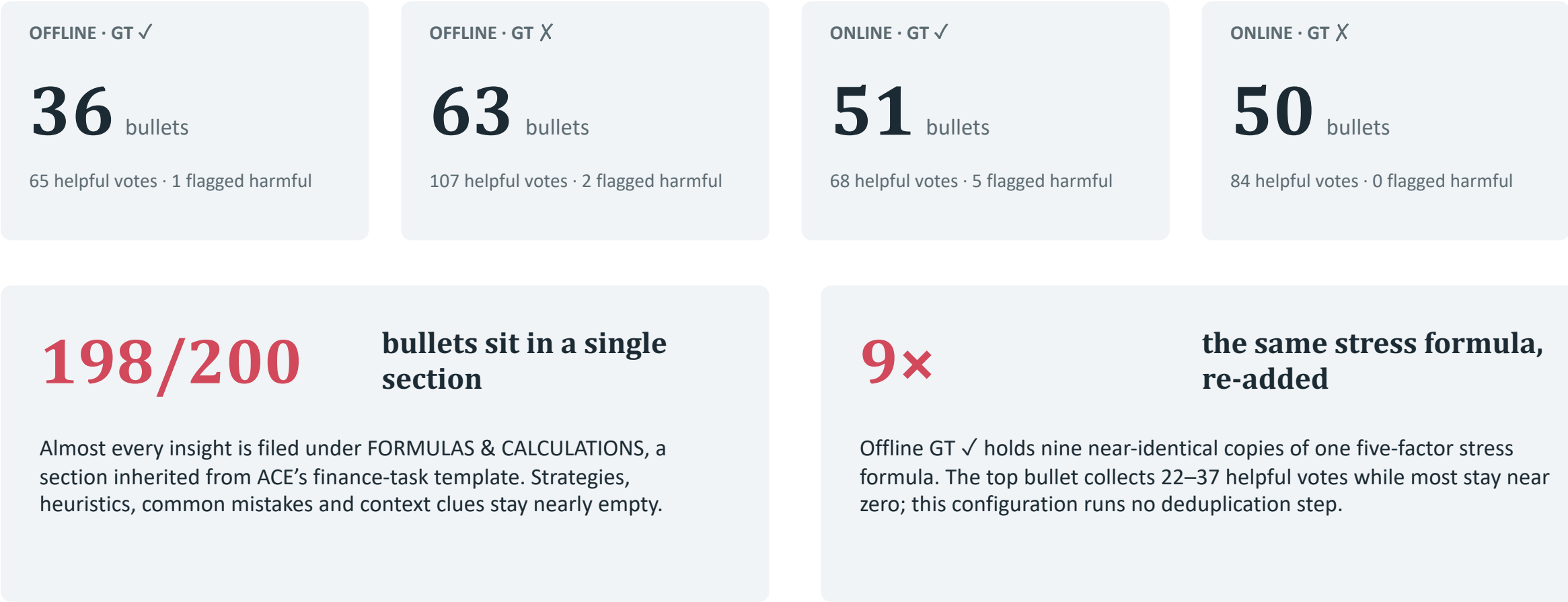
What stands out

- 1 ACE (all four conditions, +7.0 to +11.3 pp) and GEPA (+9.7 pp) clearly beat Base. GEPA sits between the ACE GT ✓ and GT X conditions.
- 2 DC (−5.0, −6.3) and ICL (−2.3) fall below Base. MIPROv2 (+1.3) sits inside its seed spread (65–74).
- 3 Conformability is high for most methods (83–98%). Offline ACE GT ✓ is the outlier: 90 / 54 / 84 across seeds.
- 4 Caveats: the judge is the same Qwen3-8B (provisional, uncalibrated); seeds reuse the same 100 posts; two seeds with invalid outputs use all-attempt F1.

Conformability: source-grounding pass rate (all four dimensions = 2) from a method-blind judge; same-family evaluation. Highlighted rows are ACE.

What ACE learned on Dreaddit: four playbooks (seed 42)

Final playbooks after one pass. Offline: 100 training posts, frozen before test. Online: 100 test posts, updated after each committed prediction.



helpful / harmful are ACE’s per-bullet counters, updated from Reflector feedback. Seed 42 shown for every condition, fixed before viewing.

Offline ACE with GT: the model invents a stress score

Playbook learned from 100 labeled training posts, then frozen for the test posts

[calc-00001] helpful 26 · harmful 0

Stress Level Calculation: Stress Level = (Uncertainty Score + Seeking Advice Score + Decision Complexity Score) / 3. If the result is ≤ 1.5 , classify as 0 (no stress); if ≥ 2.5 , classify as 1 (stress).

The most-used bullet: an average of three hand-named scores with fixed thresholds. Nothing in the task asked for a numeric score; the formula section invited one.

...then keeps re-deriving it

[calc-00007] helpful 11 · harmful 0

Stress Level = (Uncertainty Score + Seeking Advice Score + Decision Complexity Score + Emotional Language Intensity Score + Contextual Stress Indicator Score) / 5. If the ...

[calc-00013] helpful 5 · harmful 0

Stress Level = (Uncertainty Score + Seeking Advice Score + Decision Complexity Score + Emotional Language Intensity Score + Contextual Stress Indicator Score) / 5. If the ...

[calc-00024] helpful 4 · harmful 0

Stress Level = (Uncertainty Score + Seeking Advice Score + Decision Complexity Score + Emotional Language Intensity Score + Contextual Stress Indicator Score) / 5. If the ...

[ctx-00002] **helpful 8 · harmful 10 (flagged)**

Identify stress through emotional language and contextual cues, such as phrases like “...” or “...”, which indicate emotional distress and anxiety.

Offline ACE without GT: qualitative cues instead of formulas

Playbook learned from the same 100 training posts with no labels, only the model's own reflection

[calc-00001] helpful 37 · harmful 0

Identify indirect signs of stress such as uncertainty, indecision, and seeking advice. These can be strong indicators of stress even if the author does not explicitly state it.

[calc-00016] helpful 20 · harmful 0

When the author explicitly expresses self-blame and self-hatred, prioritize these as the strongest indicators of stress, as they directly reflect emotional distress and align with the playbook's guidance to focus on the most direct and emotionally charged quotes.

[calc-00017] helpful 20 · harmful 0

When the author explicitly expresses emotional distress (e.g., self-criticism, pain, crying in pain), prioritize these as the strongest indicators of stress, as they directly reflect the author's emotional state and align with the playbook's guidance to focus on the most direct and ...

[calc-00002] **helpful 3 · harmful 6 (flagged)**

Consider both direct and indirect signs of stress, and evaluate the context of the text more thoroughly. Be aware that the presence of indirect signs may not be sufficient to label the text as stress without further contextual analysis.

Online ACE: playbooks after 100 test posts, with and without GT

Updated after each committed prediction; GT ✓ reveals the label only after the prediction is saved

GT ✓ · label revealed after each prediction

[calc-00015] helpful 22 · harmful 3 (flagged)

Use a binary classification system where the presence of any explicit stress indicators, including events that 'blindsided' the author or indicate lack of preparedness, is sufficient to classify the text as 'stress'. This approach prioritizes the ...

[calc-00001] helpful 17 · harmful 0

Use a weighted scoring system where emotional indicators such as frustration, anger, and sensory overload are assigned higher weights (e.g., 3 points each) compared to other indicators (e.g., 1 point). Sum the scores and classify as stress if the ...

[calc-00019] helpful 8 · harmful 11 (flagged)

Use a binary classification system where the presence of any explicit stress indicators such as frustration, fatigue, anger, or sensory overload is sufficient to classify the text as 'stress'. The absence of these indicators, even if other emotional ...

GT ✗ · no labels, self-reflection only

[calc-00019] helpful 25 · harmful 0

Ensure that the final label always matches the reasoning and that the evidence supports the label. If multiple stress indicators are present, classify as 'stress' (label 1) regardless of the author's denial or minimization.

[ctx-00001] helpful 15 · harmful 0

Always reference playbook bullets when available to ensure that the reasoning is aligned with the expected framework and to improve the accuracy of the answer.

[calc-00014] helpful 9 · harmful 0

Prioritize the presence of multiple stress indicators over any denial or minimization by the author. If multiple stress indicators are present, classify as 'stress' (label 1) regardless of the author's denial.